

Chapter 7 Linear motion

Section 7.1 Displacement, speed and velocity

Worked example: Try yourself 7.1.1

AVERAGE VELOCITY AND CONVERTING UNITS

Sally is an athlete performing a training routine by running back and forth along a straight stretch of running track. She jogs 100m west in a time of 20s, then turns and walks 160m east in a further 45s before stopping.

a What is Sally's average velocity in m s^{-1} ?	
Thinking	Working
Calculate the displacement. Remember that total displacement is the sum of individual displacements. Sally's total journey consists of two displacements: 100m west and 160m east. Take east to be the positive direction.	$s = \text{sum of displacements}$ $= 100\text{ m west} + 160\text{ m east}$ $= -100 + 160$ $= +60\text{ m or }60\text{ m east}$
Work out the total time taken for the journey.	Time taken $= 20 + 45 = 65\text{ s}$
Substitute the values into the velocity equation.	Displacement, s , is 60m east. Time taken, t , is 65s. Average velocity $v_{\text{av}} = \frac{s}{t}$ $= \frac{60}{65}$ $= 0.92\text{ m s}^{-1}$
Velocity is a vector, so a direction must be given.	0.92 m s^{-1} east

b What is the magnitude of Sally's average velocity in km h^{-1} ?	
Thinking	Working
Convert from m s^{-1} to km h^{-1} by multiplying by 3.6.	$v_{\text{av}} = 0.92\text{ m s}^{-1}$ $= 0.92 \times 3.6$ $= 3.3\text{ km h}^{-1}$ east
As the magnitude of the velocity is needed, the direction is not required in this answer.	Magnitude of $v_{\text{av}} = 3.3\text{ km h}^{-1}$

c What is Sally's average speed in m s^{-1} ?	
Thinking	Working
Calculate the distance. Remember that distance is the length of the path covered in the entire journey. The direction does not matter. Sally travels 100m in one direction and then 160m in the other direction.	$s = 100 + 160$ $= 260\text{ m}$
Work out the total time taken for the journey.	$20 + 45 = 65\text{ s}$
Substitute the values into the speed equation.	Distance, s , is 260m. Time taken, t , is 65s. Average speed $v_{\text{av}} = \frac{s}{t}$ $= \frac{260}{65}$ $= 4.0\text{ m s}^{-1}$

d What is Sally's average speed in km h^{-1} ?

Thinking

Convert from m s^{-1} to km h^{-1} by multiplying by 3.6.

Working

$$\begin{aligned}\text{Average speed } v_{av} &= 4.0 \text{ m s}^{-1} \\ &= 4.0 \times 3.6 \\ &= 14.4 \text{ km h}^{-1}\end{aligned}$$

Section 7.1 Review

KEY QUESTIONS SOLUTIONS

1 a average speed $v_{av} = \frac{\text{distance travelled}}{\text{time taken}} = \frac{s}{t} = \frac{400}{2 \times 60} = \frac{400}{120}$
 $= 3.33 \text{ m s}^{-1}$

b average velocity $v_{av} = \frac{\text{displacement}}{\text{time taken}} = \frac{s}{t} = \frac{0}{120}$
 $= 0.00 \text{ m s}^{-1}$

Her displacement is zero because the start and finish points are the same.

2 B and C. The distance travelled is $25 \times 10 = 250 \text{ m}$, but the displacement is zero because the swimmer starts and ends at the same place.

3 a Displacement = final position – initial position
 $= 40 - 0$
 $= +40 \text{ cm}$

Distance travelled = 40 cm

b Displacement = final position – initial position
 $= 40 - 50$
 $= -10 \text{ cm}$

Distance travelled = 10 cm

c Displacement = final position – initial position
 $= 70 - 50$
 $= 20 \text{ cm}$

Distance travelled = 20 cm

d Displacement = final position – initial position
 $= 70 - 50$
 $= 20 \text{ cm}$

Distance covered = $50 + 30$
 $= 80 \text{ cm}$

4 a $s = 50 + 30 = 80 \text{ km}$

b $s = 50 \text{ km north} + 30 \text{ km south}$
 $= 50 + (-30)$
 $= 50 - 30$
 $= +20 \text{ km or } 20 \text{ km north}$

5 a The basement is 10 m downwards from the starting position on the ground floor. This can be calculated using the following equation:

$s = \text{final position} - \text{initial position}$
 $= -10 - 0$
 $= -10 \text{ m or } 10 \text{ m downwards}$

b The total displacement from the basement to the top floor is 60 m upwards. This can be calculated using the following equation:

$s = \text{final position} - \text{initial position}$
 $= +50 - (-10)$
 $= 50 + 10$
 $= +60 \text{ m or } 60 \text{ m upwards}$

- c** The total distance travelled is 70 m.

$$10 + 10 + 50 = 70 \text{ m}$$

- d** The top floor is 50 m upward from the starting position on the ground floor. This can be calculated using the following equation:

$$s = \text{final position} - \text{initial position}$$

$$= 50 - 0$$

$$= 50 \text{ m or } 50 \text{ m upwards}$$

- 6 a** average speed $v_{av} = \frac{\text{distance travelled}}{\text{time taken}}$

$$= \frac{400}{12}$$

$$= 33 \text{ m s}^{-1}$$

- b** The car travelled 25 m. This can be calculated using the following method:

$$\text{average speed } v_{av} = \frac{\text{distance travelled}}{\text{time taken}}$$

$$s = v_{av} \times t$$

$$= 33 \times 0.75$$

$$= 25 \text{ m}$$

- 7 a** $90 \text{ min} = \frac{90}{60}$

$$= 1.5 \text{ h}$$

$$\text{average speed } v_{av} = \frac{\text{distance travelled}}{\text{time taken}}$$

$$= \frac{25}{1.5}$$

$$= 17 \text{ km h}^{-1}$$

- b** To convert from km h^{-1} to m s^{-1} , you need to divide by 3.6. So:

$$\text{average speed } v_{av} = \frac{17}{3.6}$$

$$= 4.7 \text{ m s}^{-1}$$

- 8 a** average speed $v_{av} = \frac{\text{distance travelled}}{\text{time taken}}$

$$= \frac{s}{t}$$

$$= \frac{9}{10}$$

$$= 0.9 \text{ m s}^{-1}$$

- b** Displacement is 1 m east of the starting position.

$$\text{average velocity } v_{av} = \frac{\text{displacement}}{\text{time taken}}$$

$$= \frac{s}{t}$$

$$= \frac{1}{10}$$

$$= 0.1 \text{ m s}^{-1} \text{ east}$$

- 9 a** average speed $v_{av} = \frac{\text{distance travelled}}{\text{time taken}}$

$$= \frac{2.5}{0.25}$$

$$= 10 \text{ km h}^{-1}$$

- b** average velocity $v_{av} = \frac{10}{3.6} = 2.8 \text{ m s}^{-1} \text{ south}$

- 10 a** Distance travelled = 10 km north + 3 km south + x km north to finish 15 km north of the start.

$$x = 8 \text{ km north.}$$

$$\text{Total distance covered} = 10 + 3 + 8$$

$$= 21 \text{ km}$$

- b** She finishes 15 km north of her starting point. This is her displacement.

c average speed $v_{av} = \frac{\text{distance travelled}}{\text{time taken}}$

$$= \frac{21}{1.5}$$

$$= 14 \text{ km h}^{-1}$$

d average velocity $v_{av} = \frac{\text{displacement}}{\text{time taken}}$

$$= \frac{15}{1.5}$$

$$v_{av} = 10 \text{ km h}^{-1}$$

Section 7.2 Acceleration

Worked example: Try yourself 7.2.1

CHANGE IN SPEED AND VELOCITY 1

A golf ball is dropped onto a concrete floor and strikes the floor at 9.0 ms^{-1} . It then rebounds at 7.0 ms^{-1} .

a What is the change in speed of the ball?		
Thinking	Working	
Find the values for the initial speed and the final speed of the ball.	$u = 9.0 \text{ ms}^{-1}$ $v = 7.0 \text{ ms}^{-1}$	
Substitute the values into the change in speed equation: $\Delta v = v - u$	$\Delta v = v - u$ $= 7.0 - 9.0$ $= -2.0 \text{ ms}^{-1}$	Note that speed is a scalar so the negative value indicates a decrease in magnitude, as opposed to a negative direction.

b What is the change in velocity of the ball?		
Thinking	Working	
Apply the sign convention to replace the directions.	$u = 9.0 \text{ ms}^{-1}$ down $= -9.0 \text{ ms}^{-1}$ $v = 7.0 \text{ ms}^{-1}$ up $= +7.0 \text{ ms}^{-1}$	
Reverse the direction of u to get $-u$.	$u = -9.0 \text{ ms}^{-1}$ $-u = 9.0 \text{ ms}^{-1}$	
Substitute the values into the change in velocity equation: $\Delta v = v + (-u)$	$\Delta v = v + (-u)$ $= 7.0 + (+9.0)$ $= 16.0 \text{ ms}^{-1}$	
Apply the sign convention to describe the direction.	$\Delta v = 16 \text{ ms}^{-1}$ up	

Worked example: Try yourself 7.2.2

CHANGE IN SPEED AND VELOCITY 2

A golf ball is dropped onto a concrete floor and strikes the floor at 9.0 ms^{-1} . It then rebounds at 7.0 ms^{-1} . The contact time with the floor is 35 ms. What is the average acceleration of the ball during its contact with the floor?	
Thinking Note the values you will need in order to find the average acceleration (initial velocity, final velocity and time). Convert ms into s by dividing by 1000. (Note that Δv was calculated for this situation in the previous Worked example.)	Working $u = -9.0 \text{ ms}^{-1}$ $-u = 9.0 \text{ ms}^{-1}$ $v = 7.0 \text{ ms}^{-1}$ $\Delta v = v - u = 16 \text{ ms}^{-1} \text{ up}$ $t = 35 \text{ ms}$ $= 0.035 \text{ s}$
Substitute the values into the average acceleration equation.	$a = \frac{\text{change in velocity}}{\text{time taken}}$ $= \frac{\Delta v}{t} = \frac{v - u}{t}$ $= \frac{16}{0.035}$ $= 457 \text{ ms}^{-2}, \text{ which is } 460 \text{ ms}^{-2} \text{ to two significant figures}$
Acceleration is a vector, so you must include a direction in your answer.	$a = 4.6 \times 10^2 \text{ ms}^{-2} \text{ up}$

Section 7.2 Review

KEY QUESTIONS SOLUTIONS

1 $\Delta v = v - u$
 $= 3 - 10$
 $= -7$

So the change in speed is -7 km h^{-1} .

Note that speed is a scalar so the negative value indicates a decrease in magnitude, rather than a negative direction.

2 $\Delta v = v - u$
 $= 0 + (+5)$
 $= +5 \text{ ms}^{-1} \text{ or } 5 \text{ ms}^{-1} \text{ up}$

3 Down is negative, so the initial velocity is -5.0 ms^{-1} .
 $\Delta v = v - u = (3.0) - (-5.0)$
 $= +8 \text{ ms}^{-1}$
 $= 8 \text{ ms}^{-1} \text{ up}$

4 $a = \frac{\text{change in velocity}}{\text{time taken}}$
 $= \frac{v - u}{t}$
 $= \frac{0 - 7.5}{1.5}$
 $= -5.0 \text{ ms}^{-2}$
 $= 5.0 \text{ ms}^{-2} \text{ south}$

$$\begin{aligned}
 5 \quad a &= \frac{\text{change in velocity}}{\text{time taken}} = \frac{v - u}{t} \\
 &= \frac{150 - 0}{3.5} \\
 &= 43 \text{ km h}^{-1} \text{ s}^{-1}
 \end{aligned}$$

$$\begin{aligned}
 6 \quad a \quad \Delta v &= v - u \\
 &= 15 - 25 \\
 &= -10 \text{ ms}^{-1}
 \end{aligned}$$

Note that speed is a scalar so the negative value indicates a decrease in magnitude, as opposed to a negative direction.

$$\begin{aligned}
 b \quad \Delta v &= v - u \\
 &= (-15) - (25) \\
 &= -40 \text{ ms}^{-1} \\
 &= 40 \text{ ms}^{-1} \text{ west}
 \end{aligned}$$

$$\begin{aligned}
 c \quad a &= \frac{\text{change in velocity}}{\text{time taken}} \\
 &= \frac{v - u}{t} \\
 &= \frac{40}{0.050} \\
 &= 800 \text{ ms}^{-2}
 \end{aligned}$$

Magnitude only so the direction is not required.

$$\begin{aligned}
 7 \quad a \quad \Delta v &= v - u \\
 &= 8.0 - 0 \\
 &= 8.0 \text{ ms}^{-1}
 \end{aligned}$$

$$\begin{aligned}
 b \quad \Delta v &= v - u \\
 &= -8.0 - 0 \\
 &= -8.0 \text{ ms}^{-1} \\
 &= 8.0 \text{ ms}^{-1} \text{ south}
 \end{aligned}$$

$$\begin{aligned}
 c \quad a &= \frac{\text{change in velocity}}{\text{time taken}} = \frac{v - u}{t} \\
 &= \frac{8.0}{1.2} \\
 &= 6.7 \text{ ms}^{-2}
 \end{aligned}$$

$$\begin{aligned}
 8 \quad a &= \frac{v - u}{t} \\
 t &= \frac{v - u}{a} = \frac{30.0 - 10.0}{3.00} \\
 &= 6.67 \text{ s}
 \end{aligned}$$

$$\begin{aligned}
 9 \quad a &= \frac{v - u}{t} \\
 t &= \frac{v - u}{a} = \frac{00.0 - 20.0}{-2.50} \\
 &= 8.00 \text{ s}
 \end{aligned}$$

$$\begin{aligned}
 10 \quad a &= -3.00 \text{ ms}^{-2} \\
 t &= 4.00 \text{ s} \\
 v &= 0.00 \text{ ms}^{-2} \text{ (cyclist is stopped)} \\
 u &= ? \\
 v &= u + at \\
 u &= v - at = 0.00 - (-3.00) \times (4.00) \\
 &= 12.0 \text{ ms}^{-1}
 \end{aligned}$$

Section 7.3 Graphing position, velocity and acceleration over time

Worked example: Try yourself 7.3.1

ANALYSING A POSITION–TIME GRAPH

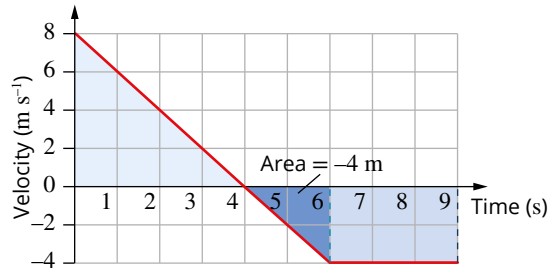
Use the graph shown in Worked example 7.3.1 to answer the following questions.

a What is the velocity of the cyclist between E and F?	
Thinking	Working
Determine the change in position (displacement) of the cyclist between E and F using: $s = \text{final position} - \text{initial position}$	At E, $x = 300 \text{ m}$. At F, $x = 0 \text{ m}$. $s = 0 - 300$ $= -300 \text{ m}$ or 300 m backwards (that is, back towards the starting point)
Determine the time taken to travel from E to F.	$\Delta t = 100 - 80$ $= 20 \text{ s}$
Calculate the gradient of the graph between E and F using: gradient of x – t graph $= \frac{\text{rise}}{\text{run}} = \frac{\Delta x}{\Delta t}$ Remember that $\Delta x = s$.	Gradient $= \frac{-300}{20}$ $= -15$
State the velocity, using: gradient of x – t graph = velocity Velocity is a vector so a direction must be given.	Since the gradient is -15 , the velocity is -15 m s^{-1} or 15 m s^{-1} backwards (towards the starting point).
b Describe the motion of the cyclist between D and E.	
Thinking	Working
Interpret the shape of the graph between D and E.	The graph is flat between D and E, indicating that the cyclist's position is not changing for this time. So the cyclist is not moving. If the cyclist is not moving, the velocity is 0 m s^{-1} .
You may confirm the result by calculating the gradient of the graph between D and E using: gradient of x – t graph $= \frac{\text{rise}}{\text{run}} = \frac{\Delta x}{\Delta t}$ Remember that $\Delta x = s$.	Gradient $= \frac{0}{20}$ $= 0$

Worked example: Try yourself 7.3.2

ANALYSING A VELOCITY–TIME GRAPH

Use the graph shown in Worked example 7.3.2 to answer the following questions.

a What is the displacement of the car from 4 to 6 s?	
Thinking Displacement is the area under the graph. So, calculate the area under the graph for the time period for which you want to find the displacement. Use displacement = $b \times h$ for squares and rectangles. Use displacement = $\frac{1}{2}(b \times h)$ for triangles.	Working  The area from 4 to 6 s is a triangle, so: $\text{area} = \frac{1}{2}(b \times h)$ $= \frac{1}{2} \times 2 \times -4$ $= -4 \text{ m}$
Displacement is a vector quantity, so a direction is needed.	displacement = 4 m west
b What is the average velocity of the car from 4 to 6 s?	
Thinking Identify the equation and variables, and apply the sign convention.	Working $v = \frac{s}{\Delta t}$ $s = -4 \text{ m}$ $\Delta t = 2 \text{ s}$
Substitute values into the equation: $v = \frac{s}{t}$	$v = \frac{s}{\Delta t}$ $= \frac{-4}{2}$ $= -2 \text{ m s}^{-1}$
Velocity is a vector quantity, so a direction is needed.	$v_{\text{av}} = 2 \text{ m s}^{-1}$ west

Worked example: Try yourself 7.3.3
FINDING ACCELERATION USING A VELOCITY–TIME GRAPH

Use the graph shown in Worked example 7.3.3 to answer the following question.
What is the acceleration of the car during the period from 4 to 6 s?

Thinking	Working
Acceleration is the gradient of a v–t graph. Calculate the gradient using: $\text{gradient} = \frac{\text{rise}}{\text{run}}$	Gradient from 4 to 6 = $\frac{\text{rise}}{\text{run}}$ $= \frac{-4}{2}$ $= -2 \text{ m s}^{-1}$
Acceleration is a vector quantity, so a direction is needed. Note: In this case, the car is moving in the negative direction and speeding up.	Acceleration = 2 m s^{-2} west.

Section 7.3 Review

KEY QUESTIONS SOLUTIONS

- D. The gradient is the displacement over the time taken, hence velocity.
- The car initially moves in a positive direction and travels 8 m in 2 s. It then stops for 2 s. The car then reverses direction for 5 s, passing back through its starting point after 8 s. It travels a further 2 m in a negative direction before stopping after 9 s.
- Reading from the graph:
 - +8 m
 - +8 m
 - +4 m
 - 2 m
- The car returns to its starting point when the position is zero again, which occurs at $t = 8$ s.
 - The velocity during the first 2 s is equal to the gradient of the graph during this interval.

$$\text{velocity} = \frac{\text{rise}}{\text{run}} = \frac{(0) - (8)}{2} = +4 \text{ m s}^{-1}$$
 - After 3 s the velocity is zero, since the gradient of the graph = 0.
 - velocity = gradient of graph = $\frac{\text{rise}}{\text{run}} = \frac{8 - 0}{4} = -2 \text{ m s}^{-1}$
 - The velocity at 8 s is -2 m s^{-1} , since the car is travelling at a constant velocity of -2 m s^{-1} between 4 s and 9 s.
 - The velocity from 8 s to 9 s = -2 m s^{-1} , since the car is travelling at a constant velocity of -2 m s^{-1} between 4 s and 9 s.
- Distance = $8 + 8 + 2 = 18$ m
 - Displacement = $\Delta x = (-2) - 0 = -2$ m
- Average speed = gradient of the line segment

$$= \frac{\text{rise}}{\text{run}}$$

$$= \frac{150}{30}$$

$$= 5 \text{ m s}^{-1}$$

- b** Average velocity = gradient of the line segment plus direction

$$\begin{aligned}
 &= \frac{\text{rise}}{\text{run}} \\
 &= \frac{200}{10} \\
 &= 20 \text{ m s}^{-1} \text{ north}
 \end{aligned}$$

The velocity is positive so the direction of the cyclist is north.

- c** Average velocity = displacement over time

$$\begin{aligned}
 &= \frac{\text{rise}}{\text{run}} \\
 &= \frac{500}{50} \\
 &= 10 \text{ m s}^{-1} \text{ north}
 \end{aligned}$$

- 8 a** Acceleration = gradient

$$\begin{aligned}
 &= \frac{\text{rise}}{\text{run}} \\
 &= 0 \text{ m s}^{-2}
 \end{aligned}$$

- b** Acceleration = gradient

$$\begin{aligned}
 &= \frac{\text{rise}}{\text{run}} \\
 &= \frac{-3}{3} \\
 &= -1 \text{ or just } 1 \text{ m s}^{-2}
 \end{aligned}$$

Magnitude only, so direction is not required.

- c** Split the area up into shapes and add the values together to get the full area under the graph.

Displacement = area under the graph

$$\begin{aligned}
 &= (b \times h) + \left(\frac{1}{2} \times b \times h \right) + \left(\frac{1}{2} \times b \times h \right) \\
 &= (4 \times 1) + \left(\frac{1}{2} \times 2 \times 2 \right) + \left(\frac{1}{2} \times 3 \times 3 \right) \\
 &= 4 + 2 + 4.5 \\
 &= 10.5 \text{ m}
 \end{aligned}$$

- d** average velocity = $\frac{\text{displacement}}{\text{time}}$

$$\begin{aligned}
 &= \frac{10.5}{7} \\
 &= 1.5 \text{ m s}^{-1}
 \end{aligned}$$

- 9 a** instantaneous velocity = gradient of the line

$$\begin{aligned}
 &= \frac{\text{rise}}{\text{run}} \\
 &= \frac{300}{15} \\
 &= 20 \text{ m s}^{-1} \text{ north}
 \end{aligned}$$

- b** instantaneous velocity = gradient of the line

$$\begin{aligned}
 &= \frac{\text{rise}}{\text{run}} \\
 &= \frac{-600}{15} \\
 &= -40 \text{ or } 40 \text{ m s}^{-1} \text{ south}
 \end{aligned}$$

- 10 a** Reading from the graph, the curve flattens out after 80 s.

- b** Draw a tangent to the graph at 10 s and determine the gradient of the tangent.

$$\begin{aligned}
 \text{gradient} &= \frac{\text{rise}}{\text{run}} \\
 &= \text{approx. } \frac{35}{50} \text{ or } \frac{53}{40} \\
 &= 1.2 \text{ or } 1.3 \text{ m s}^{-2} \text{ (answers may vary slightly)}
 \end{aligned}$$

- c** Draw a tangent to the graph at 40s and determine the gradient of the tangent.

$$\text{gradient} = \frac{\text{rise}}{\text{run}}$$

$$= \text{approx. } \frac{34}{90} \text{ or } \frac{35}{85}$$

$$= 0.38 \text{ or } 0.41 \text{ ms}^{-2} \text{ (answers may vary slightly)}$$

- d** displacement = area under the graph

There are various methods for calculating this, but counting squares gives 49 squares, each of area 10×10 .

$$49 \times 10 \times 10 = 4900 \text{ m or } 4.9 \text{ km}$$

Section 7.4 Equations for uniform acceleration

Worked example: Try yourself 7.4.1

USING THE EQUATIONS OF MOTION

A snowboarder in a race is travelling 15 ms^{-1} east as she crosses the finishing line. She then decelerates uniformly until coming to a stop over a distance of 30 m.

a What is her acceleration as she comes to a stop?	
Thinking	Working
Write down the known quantities and the quantity that you need to find. Apply the sign convention that east is positive and west is negative.	$s = +30 \text{ m}$ $u = +15 \text{ ms}^{-1}$ $v = 0 \text{ ms}^{-1}$ $a = ?$
Identify the correct equation to use.	$v^2 = u^2 + 2as$
Substitute known values into the equation and solve for a . Include units with the answer.	$v^2 = u^2 + 2as$ $0^2 = 15^2 + 2 \times a \times 30$ $a = \frac{0 - 225}{60}$ $= -3.8 \text{ ms}^{-2}$
Use the sign convention to state the answer with its direction.	$a = 3.8 \text{ ms}^{-2} \text{ west}$

b How long does she take to come to a stop?	
Thinking	Working
Write down the known quantities and the quantity you need to find. Apply the sign convention that east is positive and west is negative.	$s = 30 \text{ m}$ $u = 15 \text{ ms}^{-1}$ $v = 0 \text{ ms}^{-1}$ $a = -3.8 \text{ ms}^{-2}$ $t = ?$
Identify the correct equation to use. Since you now know four values, any equation involving t will work.	$v = u + at$
Substitute known values into the equation and solve for t . Include units with the answer.	$t = \frac{0 - 15}{-3.8}$ $= 3.9 \text{ s}$

c What is the average velocity of the snowboarder as she comes to a stop?

Thinking

Write down the known quantities and the quantity that you need to find.
Apply the sign convention that east is positive and west is negative.

Identify the correct equation to use.

Substitute known quantities into the equation and solve for v_{av} .
Include units with the answer.

Use the sign convention to state the answer with its direction.

Working

$$u = +15 \text{ ms}^{-1}$$

$$v = 0 \text{ ms}^{-1}$$

$$v_{av} = ?$$

$$v_{av} = \frac{1}{2} (u + v)$$

$$v_{av} = \frac{1}{2} (u + v)$$

$$= \frac{1}{2} (15 + 0)$$

$$= 7.5 \text{ ms}^{-1}$$

$$v_{av} = 7.5 \text{ ms}^{-1} \text{ east}$$

Section 7.4 Review

KEY QUESTIONS SOLUTIONS

1 E. The chosen equation must contain s , u , v and a .

2 a $u = 0 \text{ ms}^{-1}$, $s = 400 \text{ m}$, $t = 16 \text{ s}$, $a = ?$

$$s = ut + \frac{1}{2} at^2$$

$$400 = 0 + \frac{1}{2} a \times 16^2$$

$$a = \frac{400}{256} \times 2$$

$$= 3.1 \text{ ms}^{-2}$$

b $u = 0 \text{ ms}^{-1}$, $s = 400 \text{ m}$, $t = 16 \text{ s}$, $a = 3.1$, $v = ?$

$$v = u + at$$

$$= 0 + 3.1 \times 16$$

$$= 50 \text{ ms}^{-1}$$

c $50 \text{ ms}^{-1} \times 3.6 = 180 \text{ km h}^{-1}$

3 a $u = 0 \text{ ms}^{-1}$, $t = 8.0 \text{ s}$, $v = 16 \text{ ms}^{-1}$, $a = ?$

$$v = u + at$$

$$16 = 0 + a \times 8.0$$

$$a = \frac{16}{8.0}$$

$$= 2.0 \text{ ms}^{-2}$$

b $v_{av} = \frac{u + v}{2}$

$$= \frac{0 + 16}{2}$$

$$= 8 \text{ ms}^{-1}$$

c $u = 0 \text{ ms}^{-1}$, $t = 8.0 \text{ s}$, $v = 16 \text{ ms}^{-1}$, $a = 2.0 \text{ ms}^{-2}$, $s = ?$

$$s = \frac{1}{2} (u + v)t$$

$$= \frac{1}{2} (0 + 16) \times 8.0$$

$$= 64 \text{ m}$$

- 4 a** $u = 0 \text{ ms}^{-1}$, $v = 160 \text{ ms}^{-1}$, $t = 4.0 \text{ s}$, $a = ?$

$$v = u + at$$

$$160 = 0 + a \times 4.0$$

$$a = 40 \text{ ms}^{-2}$$

- b** In the first 4.0s: $u = 0$, $t = 4.0$, $v = 160$, $a = 40$, $s = ?$

$$s = \frac{1}{2} (u + v)t$$

$$= \frac{1}{2} (0 + 160) \times 4.0$$

$$= 80 \times 4.0$$

$$= 320 \text{ m}$$

In the last 5.0s:

$$u = 160 \text{ ms}^{-1}$$
, $t = 5.0 \text{ s}$, $v = 160 \text{ ms}^{-1}$, $a = 0 \text{ ms}^{-2}$, $s = ?$

$$s = \frac{1}{2} (u + v)t$$

$$= \frac{1}{2} (160 + 160) \times 5.0$$

$$= 160 \times 5.0$$

$$= 800 \text{ m}$$

Total distance in 9.0s:

$$= 320 + 800$$

$$= 1120 \text{ m}$$

$$= 1.12 \text{ km} = 1.1 \text{ km (to two significant figures)}$$

- c** $160 \text{ ms}^{-1} \times 3.6 = 576 \text{ km h}^{-1} = 580 \text{ km h}^{-1}$ (to two significant figures)

- d** $u = 0 \text{ ms}^{-1}$, $v = 160 \text{ ms}^{-1}$

$$v_{av} = \frac{u + v}{2}$$

$$= \frac{0 + 160}{2}$$

$$= 80 \text{ ms}^{-1}$$

e $v_{av} = \frac{s}{t}$

$$= \frac{1120}{9}$$

$$= 124.4 \text{ ms}^{-1} = 120 \text{ ms}^{-1}$$
 (to two significant figures)

- 5 a** $u = 4.2 \text{ ms}^{-1}$, $t = 0.5 \text{ s}$, $v = 6.7 \text{ ms}^{-1}$, $a = ?$

$$v = u + at$$

$$6.7 = 4.2 + a \times 0.50$$

$$a = \frac{6.7 - 4.2}{0.50}$$

$$= 5.0 \text{ ms}^{-2}$$

- b** $u = 4.2 \text{ ms}^{-1}$, $t = 0.5 \text{ s}$, $v = 6.7 \text{ ms}^{-1}$, $a = 5.0 \text{ ms}^{-2}$, $s = ?$

$$s = \frac{1}{2} (u + v)t$$

$$= \frac{1}{2} (4.2 + 6.7) \times 0.50$$

$$= 2.725 \text{ or } 2.7 \text{ m}$$

c $v_{av} = \frac{u + v}{2}$

$$= \frac{4.2 + 6.7}{2}$$

$$= 5.45 = 5.5 \text{ ms}^{-1}$$
 (to two significant figures)

- 6** D. The stone is travelling downwards, so the velocity is downwards. As the stone strikes the water, it quickly decelerates, so the acceleration is upwards.

- 7 a** $u = -28 \text{ ms}^{-1}$, $v = 0 \text{ ms}^{-1}$, $s = -4.0 \text{ m}$, $a = ?$
 $v^2 = u^2 + 2as$
 $0 = (-28)^2 + 2 \times a \times -4.0$
 $a = \frac{-784}{-8.0}$
 $= 98 \text{ ms}^{-2}$
- b** $u = -28 \text{ ms}^{-1}$, $v = 0 \text{ ms}^{-1}$, $s = -4.0 \text{ m}$, $a = 98 \text{ ms}^{-2}$, $t = ?$
 $v = u + at$
 $0 = -28 + 98t$
 $t = \frac{28}{98}$
 $= 0.29 \text{ s}$
- c** $u = -28 \text{ ms}^{-1}$, $s = -2.0 \text{ m}$, $a = 98 \text{ ms}^{-2}$, $v = ?$
 $v^2 = u^2 + 2as$
 $= (-28)^2 + 2 \times 98 \times -2.0$
 $= 784 - 392$
 $v = 19.8 = 20 \text{ ms}^{-1}$ to two significant figures
- 8 a** $u = \frac{75 \text{ kmh}^{-1}}{3.6} = 20.83 = 21 \text{ ms}^{-1}$ to two significant figures
- b** $u = 21 \text{ ms}^{-1}$, $a = 0 \text{ ms}^{-2}$, $t = 0.25 \text{ s}$, $s = ?$
 $s = ut + \frac{1}{2} at^2$
 $= 21 \times 0.25$
 $= 5.25$
 $= 5.3 \text{ m}$ to two significant figures
- c** $u = 21$, $a = -6.0$, $v = 0$, $t = ?$
 $v^2 = u^2 + 2as$
 $0 = (21)^2 + 2 \times -6.0 \times s$
 $s = \frac{(21)^2}{12}$
 $= 36.75$
 $= 37 \text{ m}$ to two significant figures
- d** $5.3 + 37 = 42.3 = 42 \text{ m}$ to two significant figures
- 9 a** $u = 0 \text{ ms}^{-1}$, $a = 2.0 \text{ ms}^{-2}$, $s = 4.0 \text{ m}$
 $v^2 = u^2 + 2as$
 $= 0 + 2(2.0 \times 4.0)$
 $v = 4.0 \text{ ms}^{-1}$
- b** $v^2 = u^2 + 2as$
 $= 0 + 2(2.0 \times 8.0)$
 $v = 5.7 \text{ ms}^{-1}$
- c** $v = u + at$
 $4.0 = 0 + 2.0t$
 $t = 2.0 \text{ s}$
- d** $v = u + at$
 $5.7 = 0 + 2.0t$
 $t = 2.85 \text{ s}$
 The time taken to travel the final 4.0 m is $2.85 \text{ s} - 2.0 \text{ s} = 0.85 \text{ s}$.
- 10 a** $v = 12 \text{ ms}^{-1}$, $a = 1.5 \text{ ms}^{-2}$, $u = 0 \text{ ms}^{-1}$
 $v = u + at$
 $12 = 0 + 1.5t$
 $t = 8.0 \text{ s}$

- b** The bus will catch up with Anna when they have each travelled the same distance from the point at which Anna first passes the bus.

Anna: constant velocity, so $s = 12 \times t$

Bus: uniform acceleration $u = 0$, $a = 1.5 \text{ ms}^{-2}$, $s = ?$, $t = ?$

$$s = ut + \frac{1}{2} at^2$$

$$= 0.75t^2$$

When the bus catches up with Anna, their displacements are equal, so:

$$12t = 0.75t^2$$

$$t = 16 \text{ s}$$

- c** $s = 12 \times 16 = 192 \text{ m}$

Section 7.5 Vertical motion

Worked example: Try yourself 7.5.1

VERTICAL MOTION

A construction worker accidentally knocks a hammer from a building so that it falls vertically a distance of 60 m to the ground. Use $g = -9.80 \text{ ms}^{-2}$ and ignore air resistance when answering these questions.

a How long does the hammer take to fall halfway, to 30 m?	
Thinking	Working
Write down the known quantities and the quantity that you need to find. Apply the sign convention that up is positive and down is negative.	$s = -30 \text{ m}$ $u = 0 \text{ ms}^{-1}$ $a = -9.80 \text{ ms}^{-2}$ $t = ?$
Identify the correct equation for uniform acceleration to use.	$s = ut + \frac{1}{2} at^2$
Substitute known values into the equation and solve for t . Think about whether the value seems reasonable.	$-30 = 0 \times t + \frac{1}{2} \times -9.80 \times t^2$ $-30 = -4.90t^2$ $t = \sqrt{\frac{-30}{-4.90}}$ $= 2.5 \text{ s}$
b How long does it take the hammer to fall all the way to the ground?	
Thinking	Working
Write down the known quantities and the quantity that you need to find. Apply the sign convention that up is positive and down is negative.	$s = -60 \text{ m}$ $u = 0 \text{ ms}^{-1}$ $a = -9.80 \text{ ms}^{-2}$ $t = ?$
Identify the correct equation for uniform acceleration to use.	$s = ut + \frac{1}{2} at^2$
Substitute known values into the equation and solve for t . Think about whether the value seems reasonable. Notice that the hammer takes 2.5 s to travel the first 30 m and only 1.0 s to travel the final 30 m. This is because it is accelerating.	$-60 = 0 \times t + \frac{1}{2} \times -9.80 \times t^2$ $-60 = -4.90t^2$ $t = \sqrt{\frac{-60}{-4.90}}$ $= 3.5 \text{ s}$

c What is the velocity of the hammer as it hits the ground?

Thinking	Working
Write down the known quantities and the quantity that you need to find. Apply the sign convention that up is positive and down is negative.	$s = -60\text{ m}$ $u = 0\text{ ms}^{-1}$ $v = ?$ $a = -9.80\text{ ms}^{-1}$ $t = 3.5\text{ s}$
Identify the correct equation to use. Since you now know four values, any equation involving v will work.	$v = u + at$
Substitute the known values into the equation and solve for v . Think about whether the value seems reasonable.	$v = 0 + (-9.80) \times 3.5$ $= -34\text{ ms}^{-1}$
Use the sign and direction convention to describe the direction of the final velocity.	$v = -34\text{ ms}^{-1}$ or 34 ms^{-1} downwards

Worked example: Try yourself 7.5.2

MAXIMUM HEIGHT PROBLEMS

On winning a cricket match, a fielder throws a cricket ball vertically into the air at 15 ms^{-1} . In the following questions, ignore air resistance and use $g = 9.80\text{ ms}^{-2}$.

a Determine the maximum height reached by the ball.

Thinking	Working
Write down the known quantities and the quantity that you need to find. At the maximum height the velocity is zero. Apply the sign convention that up is positive and down is negative.	$u = 15\text{ ms}^{-1}$ $v = 0$ $a = -9.80\text{ ms}^{-2}$ $s = ?$
Identify the correct equation to use.	$v^2 = u^2 + 2as$
Substitute known values into the equation and solve for s .	$0 = (15)^2 + 2 \times (-9.80) \times s$ $s = \frac{-225}{-19.6}$ $\therefore s = +11.5\text{ m}$, i.e. the ball reaches a height of 11.5 m .

b Calculate the time that the ball takes to return to its starting position.

Thinking	Working
To work out the time the ball is in the air, first calculate the time it takes to reach its maximum height. Write down the known quantities and the quantity that you need to find.	$u = 15\text{ ms}^{-1}$ $v = 0\text{ ms}^{-1}$ $a = -9.80\text{ ms}^{-2}$ $s = 11.5\text{ m}$ $t = ?$
Identify the correct equation to use.	$v = u + at$
Substitute known values into the equation and solve for t .	$0 = 15 + (-9.80 \times t)$ $9.80t = 15$ $\therefore t = 1.5\text{ s}$ The ball takes 1.5 s to reach its maximum height. It will therefore take 1.5 s to fall from this height back to its starting point and so it takes 3.0 s to return to its starting position.

Section 7.5 Review

KEY QUESTIONS SOLUTIONS

- 1 The upwards velocity will decrease by 9.80 ms^{-1} every second until the ball stops at its highest point. The velocity will then increase by 9.80 ms^{-1} every second downwards until it hits the ground.
- 2 B. The acceleration of a falling object is due to gravity, so it is constant.
- 3 A and D. Acceleration due to gravity is constant (down), however, velocity changes throughout the journey as it is zero at the top of the flight.
- 4 **a** $u = 0 \text{ ms}^{-1}$, $a = -9.80 \text{ ms}^{-2}$, $t = 3 \text{ s}$, $v = ?$

$$v = u + at$$

$$= 0 + (-9.80) \times 3.0$$

$$= 29 \text{ ms}^{-1} \text{ (no direction required for speed)}$$
b $s = -30 \text{ m}$, $u = 0 \text{ ms}^{-1}$, $a = -9.80 \text{ ms}^{-2}$, $v = ?$

$$v^2 = u^2 + 2as$$

$$= 0 + 2 \times (-9.80) \times (-30)$$

$$v = \sqrt{588}$$

$$= 24 \text{ ms}^{-1}$$
c $v_{\text{av}} = \frac{1}{2} (u + v)$

$$= \frac{1}{2} (0 + 24)$$

$$= 12 \text{ ms}^{-1} \text{ (down)}$$
- 5 **a** The same as. The acceleration of a falling object is due to gravity, so it is constant no matter the direction of vertical travel (upwards or downwards).
b The same as. The flight is symmetrical, so the starting and landing speeds are the same, but in opposite directions.
- 6 **a** $v = 0 \text{ ms}^{-1}$, $a = -9.80 \text{ ms}^{-2}$, $t = 1.5 \text{ s}$, $u = ?$

$$v = u + at$$

$$0 = u - 9.80 \times 1.5$$

$$u = 14.7$$

$$= 15 \text{ ms}^{-1} \text{ (to two significant figures)}$$
b $u = 15 \text{ ms}^{-1}$, $v = 0 \text{ ms}^{-1}$, $a = -9.80 \text{ ms}^{-2}$, $t = 1.5 \text{ s}$, $s = ?$

$$s = \frac{1}{2} (u + v)t$$

$$= \frac{1}{2} (15 + 0) \times 1.5$$

$$= 11.25$$

$$= 11 \text{ m (to two significant figures)}$$
- 7 **a** $u = 0 \text{ ms}^{-1}$, $a = -9.80 \text{ ms}^{-2}$, $t = 0.40 \text{ s}$, $v = ?$

$$v = u + at$$

$$= 0 - 9.80 \times 0.40$$

$$= -3.92$$

$$= -3.9 \text{ ms}^{-1} \text{ (to two significant figures)}$$
b $u = 0 \text{ ms}^{-1}$, $a = -9.80 \text{ ms}^{-2}$, $t = 0.40 \text{ s}$, $v = -3.9 \text{ ms}^{-1}$, $s = ?$

$$s = \frac{1}{2} (u + v)t$$

$$= \frac{1}{2} (0 + (-3.9)) \times 0.40$$

$$= -0.78$$

$$= 0.78 \text{ m}$$

c $u = 0, a = -9.80, t = 0.20, s = ?$

$$s = ut + \frac{1}{2} at^2$$

$$= 0 + \frac{1}{2} \times -9.80 \times (0.20)^2$$

$$= -0.20$$

$$= 0.20\text{m}$$

d Distance in last $0.20\text{s} = 0.78 - 0.20$
 $= 0.58\text{m}$

8 a The time to the top is half of the total time, i.e. 2.0s .

b $v = 0\text{ms}^{-1}, a = -9.80\text{ms}^{-2}, t = 2\text{s}, u = ?$

$$v = u + at$$

$$0 = u + -9.80 \times 2$$

$$u = 9.80 \times 2$$

$$= 19.6 \text{ or } 20\text{ms}^{-1}$$

c $v = 0, a = -9.80, t = 2, u = 19.6, s = ?$

$$s = vt + \frac{1}{2} at^2$$

$$= 0 + \frac{1}{2} \times -9.80 \times (2)^2$$

$$= 19.6$$

$$= 20\text{m (to two significant figures)}$$

d The lid returns to its starting position, so the final velocity will be same as the launch velocity, but in the opposite direction, i.e. 20ms^{-1} downwards.

9 a Shot-put: $u = 0\text{ms}^{-1}, a = -9.80\text{ms}^{-2}, s = -60.0\text{m}, t = ?$

$$s = ut + \frac{1}{2} at^2$$

$$-60.0 = 0 + \frac{1}{2} \times -9.80 \times t^2$$

$$t^2 = \frac{60}{\frac{1}{2} \times 9.80}$$

$$t = 3.5\text{s}$$

b $100\text{g mass: } u = -10.0\text{ms}^{-1}, a = -9.80\text{ms}^{-2}, s = -70.0\text{m}, v = ?, t = ?$

$$v^2 = u^2 + 2as$$

$$= (-10.0)^2 + 2 \times (-9.80) \times (-70.0)$$

$$= 1472$$

$$v = \pm 38.4\text{ms}^{-1}$$

Because the mass has a downwards velocity, we use the negative value.

$$v = -38.4\text{ms}^{-1}$$

$$v = u + at$$

$$-38.4 = -10.0 - 9.80t$$

$$9.80t = -10.0 + 38.4$$

$$t = 2.9\text{s}$$

You can also solve this using the formula $s = ut + \frac{1}{2} at^2$ and the quadratic formula.

10 a $s = vt - \frac{1}{2} at^2$

$$15.0 = 0 - 0.5 \times 9.80 \times t^2$$

$$t = 1.7\text{s}$$

b From maximum height of 15.0m , the ball will fall by 11.0m . Find how long it takes to travel this 11.0m .

$$s = ut - \frac{1}{2} at^2$$

$$-11.0 = 0 + 0.5 \times (-9.80) \times t^2$$

$$t = 1.5\text{s}$$

$$\text{Total time from bounce} = 1.7 + 1.5 = 3.2\text{s}$$

- 11 a** $u = 8.00 \text{ ms}^{-1}$, $a = -9.80 \text{ ms}^{-2}$, $t = 3.00 \text{ s}$

It is at its maximum height when $v = 0$.

$$v^2 = u^2 + 2as$$

$$s = \frac{v^2 - u^2}{2a}$$

$$= \frac{0 - 8.00^2}{2 \times (-9.80)}$$

$$= 3.27 \text{ m}$$

- b** when $v = 0$

$$v = u + at$$

$$t = \frac{v - u}{a} = \frac{0 - 8.00}{-9.80}$$

$$= 0.816 \text{ s}$$

- c** time taken to fall = 3.00 s

$$s = ?$$

$$s = ut + \frac{1}{2}at^2$$

$$= 8.00 \times 3.00 + \frac{1}{2} \times (-9.80) \times (3.00)^2$$

$$s = -20.1 \text{ m}$$

height of cliff is 20.1 m above the sea

CHAPTER 7 REVIEW

1 $\frac{95}{3.6} = 26 \text{ ms}^{-1}$

2 $15 \times 3.6 = 54 \text{ km h}^{-1}$

3 average speed = $\frac{\text{distance}}{\text{time}}$

$$= \frac{15 + 5 + 5 + 5}{3.0}$$

$$= 10 \text{ km h}^{-1}$$

4 a average velocity = $\frac{\text{displacement}}{\text{time}}$

$$= \frac{20}{3.0}$$

$$= 6.7 \text{ km h}^{-1} \text{ north}$$

b $6.7 \text{ km h}^{-1} \text{ north} = \frac{6.7}{3.6}$

$$= 1.9 \text{ ms}^{-1} \text{ north}$$

5 $\Delta v = 4.0 - 6.0$

$$= -2.0 \text{ ms}^{-1}$$

The change in speed is -2.0 ms^{-1} . That is, it has decreased by 2.0 ms^{-1} . Speed is a scalar and has no direction.

- 6** B. The car is moving in a positive direction so its velocity is positive. The car is slowing down so its acceleration is negative.

7 $a = \frac{v - u}{t}$

$$= \frac{-15}{2.5}$$

$$= -6 \text{ ms}^{-2}$$

or

$$u = 15 \text{ ms}^{-1}, v = 0 \text{ ms}^{-1}, t = 2.5 \text{ s}, a = ?$$

$$v = u + at$$

$$0 = 15 + a \times 2.5$$

$$a = \frac{-15}{2.5}$$

$$= -6 \text{ ms}^{-2}$$

- 8**
- a** The only positive gradient section is from 10 to 25 s.
 - b** The only negative gradient section is from 30 to 45 s.
 - c** The motorbike is stationary when the sections on the position–time graph are horizontal. The horizontal sections are from 0 to 10 s, from 25 to 30 s and from 45 to 60 s.
 - d** The zero position is at 42.5 s or 43 s.
- 9**
- a** Graph B is the correct answer as it shows speed decreasing to zero to show the car stopping.
 - b** Graph A is the correct graph because it shows a constant value for speed. This is indicated by a straight horizontal line on a velocity–time graph.
 - c** Graph C is the correct graph because it shows velocity increasing from zero in a straight line, indicating uniform acceleration.
- 10**
- a** Displacement is the area under a velocity–time graph. Area can be determined by counting squares under the graph, then multiplying by the area of each square. This gives approximately 57 squares $\times (2 \times 1) = 114$ m. Alternatively, you can break the area into various shapes and find the sum of their areas:
 $72 + 14 + 18 + 10 = 114$ m.
 The result is positive, which means the displacement is north of the starting point.
 The cyclist's displacement is 114 m north.
 - b** Average velocity = $\frac{\text{displacement}}{\text{time}}$

$$= \frac{114}{11.0}$$

$$= 10.4 \text{ m s}^{-1}$$
 - c** Acceleration is the gradient of the graph. At $t = 1$ s, the gradient is flat and therefore zero. This could also be calculated as follows:

$$\text{gradient} = \frac{\text{rise}}{\text{run}}$$

$$= 0 \text{ m s}^{-2}$$
 - d** Acceleration at $t = 10$ s is:

$$\text{gradient} = \frac{\text{rise}}{\text{run}}$$

$$= -\frac{14}{2}$$

$$= -7 \text{ or } 7 \text{ m s}^{-2} \text{ south}$$
 - e** A. The velocity is always positive (or zero) indicating that the cyclist only travelled in one direction.
- 11** $u = 0 \text{ m s}^{-1}$, $a = 3.5 \text{ m s}^{-2}$, $t = 4.5 \text{ s}$, $v = ?$

$$v = u + at$$

$$= 0 + 3.5 \times 4.5$$

$$= 15.75$$

$$= 16 \text{ m s}^{-1} \text{ (to two significant figures)}$$
- 12**
- a** $u = 0 \text{ m s}^{-1}$, $s = 2 \text{ m}$, $t = 1 \text{ s}$, $a = ?$

$$s = ut + \frac{1}{2} at^2$$

$$2.0 = 0 + \frac{1}{2} \times a \times (1.0)^2$$

$$a = 4.0 \text{ m s}^{-2}$$
 - b** $u = 0 \text{ m s}^{-1}$, $t = 1 \text{ s}$, $a = 4 \text{ m s}^{-2}$, $v = ?$

$$v = u + at$$

$$= 0 + 4.0 \times 1.0$$

$$= 4.0 \text{ m s}^{-1}$$
 - c** After 2.0 s the total distance travelled:
 $u = 0 \text{ m s}^{-1}$, $t = 2 \text{ s}$, $a = 4 \text{ m s}^{-2}$, $s = ?$

$$s = ut + \frac{1}{2} at^2$$

$$= 0 + 0.5 \times 4.0 \times (2.0)^2$$

$$= 8.0 \text{ m}$$
 Distance travelled during the 2nd second = $8.0 - 2.0 = 6.0 \text{ m}$.

- 13 a** $u = 10 \text{ ms}^{-1}$, $v = 0 \text{ ms}^{-1}$, $s = 10 \text{ m}$, $a = ?$

$$v^2 = u^2 + 2as$$

$$0 = 10^2 + 2 \times a \times 10$$

$$a = -\frac{100}{20}$$

$$= -5.0 \text{ ms}^{-2}$$

- b** $u = 10 \text{ ms}^{-1}$, $v = 0 \text{ ms}^{-1}$, $s = 10 \text{ m}$, $a = -5 \text{ ms}^{-2}$, $t = ?$

$$v = u + at$$

$$0 = 10 - 5t$$

$$t = 2.0 \text{ s}$$

- 14 a** She starts at $+4 \text{ m}$.

- b** She is at rest during section A and C.

- c** She is moving in a positive direction during section B with a velocity $+0.8 \text{ ms}^{-1}$.

- d** She is moving in the negative direction at 2.4 ms^{-1} during section D.

- e** $d = 8 + 12$

$$= 20 \text{ m}$$

$$t = 25 \text{ s}$$

$$v_{av} = \frac{s}{t}$$

$$= \frac{20}{25}$$

$$= 0.8 \text{ ms}^{-1}$$

- 15 a** acceleration = gradient

$$= \frac{\text{rise}}{\text{run}}$$

$$= \frac{8}{4}$$

$$= 2 \text{ ms}^{-1}$$

- b** The bus will overtake the bike when they have both travelled the same distance, given by the areas under the two graphs. After 8 s, the bus has travelled 56 m and the bike 64 m. After 10 s, the bus has travelled 80 m and the bike 80 m.

Algebraically, this could be determined by:

The displacement for the bus = $56 + 12(t - 8)$

The displacement for the bike = $8t$

Equating these two displacements gives:

$$8t = 56 + 12t - 96$$

$$12t - 8t = 96 - 56$$

$$4t = 40$$

$$t = 10 \text{ s}$$

- c** After 10 s the bike has travelled $10 \times 8 = 80 \text{ m}$.

- d** average velocity $v_{av} = \frac{\text{displacement}}{\text{time taken}}$

$$\text{Displacement} = \left(\frac{1}{2} \times 4 \times 8\right) + (4 \times 8) + \left(\frac{1}{2} \times 4 \times 4\right)$$

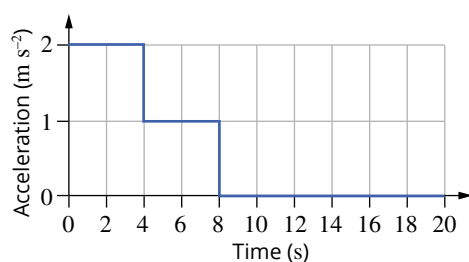
$$= 16 + 32 + 8$$

$$= 56 \text{ m}$$

$$\text{So } v_{av} = \frac{56}{8}$$

$$= 7 \text{ ms}^{-1}$$

- 16 a**



- b** The change in velocity of the bus over the first 8 s is determined by calculating the area under the acceleration–time graph from $t = 0$ to $t = 8$ s, i.e. $+12 \text{ ms}^{-1}$.
- 17** The marble slows down by 9.80 ms^{-1} each second so it will take 4 s to stop momentarily at the top of its motion. It has a positive velocity that changes to zero on the way up. Its acceleration is constant at -9.80 ms^{-2} due to gravity.
- 18** D. The acceleration of a falling object is due to gravity, so it is constant.
- 19** B. Initial velocity is upwards, it is zero at the top and downwards on the way back down. Acceleration due to gravity is always downwards.
- 20 a** The area under the v – t graph up to 3 s gives:

$$s = \frac{1}{2} \times 3 \times 30$$

$$= 45 \text{ m}$$

or

$$u = 30, v = 0, t = 3, s = ?$$

$$s = \frac{1}{2} (u + v)t$$

$$= \frac{1}{2} (30 + 0) \times 3$$

$$= 45 \text{ m}$$

- b** From the graph, the ball goes up for 3 s then down for 3 s, giving a total time of 6 s, or:

$$u = 30 \text{ ms}^{-1}, v = -30 \text{ ms}^{-1}, a = -10 \text{ ms}^{-2}, t = ?$$

$$v = u + at$$

$$-30 = 30 - 10t$$

$$t = \frac{60}{10}$$

$$= 6 \text{ s}$$

- c** From the v – t graph, the velocity at $t = 5$ s is -20 or 20 ms^{-1} down, or:

$$u = 30 \text{ ms}^{-1}, a = -10 \text{ ms}^{-2}, t = 5 \text{ s}, v = ?$$

$$v = u + at$$

$$= 30 + (-10) \times 5$$

$$= 30 - 50$$

$$= -20 \text{ ms}^{-1}$$

A negative value indicates down, therefore the correct answer is 20 ms^{-1} down.

- d** Acceleration is always 10 ms^{-2} down.

- 21 a** Balloon: $u = -8.0 \text{ ms}^{-1}, a = 0 \text{ ms}^{-2}, s = -80 \text{ m}, t = ?$

The balloon has constant speed. Use $v_{av} = \frac{s}{t}$ so:

$$t = \frac{s}{v_{av}}$$

$$= \frac{80}{8.0}$$

$$= 10 \text{ s}$$

- b** Coin: $u = -8.0 \text{ ms}^{-1}, a = -9.80 \text{ ms}^{-2}, s = -80 \text{ m}, v = ?$

$$v^2 = u^2 + 2as$$

$$= (-8)^2 + 2 \times -9.80 \times -80$$

$$= 64 + 1568$$

$$v = 40.4$$

$$= 40 \text{ ms}^{-1} \text{ to two significant figures}$$

- c** Coin: $u = -8.0, a = -9.80, s = -80, v = -40.4, t = ?$

$$v = u + at$$

$$-40.4 = -8.0 - 9.80t$$

$$9.80t = -8.0 + 40.4$$

$$t = 3.3 \text{ s}$$

Balloon takes 10 s to land, coin takes 3.3 s, so $10 - 3.3 = 6.7 \text{ s}$ difference.

22 $t = 1.5\text{ s}$, $v = 0\text{ ms}^{-1}$, $a = -9.80\text{ ms}^{-2}$, $u = ?$

$$v = u + at$$

$$0 = u + (-9.80 \times 1.5)$$

$$u = 14.7$$

$$u = 15\text{ ms}^{-1} \text{ up}$$

23 $t = 1.5\text{ s}$, $v = 0\text{ ms}^{-1}$, $a = -9.80\text{ ms}^{-2}$, $u = 14.7\text{ ms}^{-1}$, $s = ?$

$$v^2 = u^2 + 2as$$

$$0 = (14.7)^2 + 2 \times -9.80 \times s$$

$$s = 11\text{ m}$$